

CMP-11 - Temporal Service

Provide bounded time, ordering, freshness, lease, deadline, temporal commitment, and causality services without treating wall-clock time as authoritative distributed ordering state.

Authority and state ownership

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 13. Authoritative state: Time Facts, Logical clocks, lease records, time-source assessments, Temporal Commitment Records, scheduling constraints, deadline and freshness policies. Explicitly non-owned: work placement, event ownership, business scheduling authority, authority decisions.

Critical invariant

Ordering and freshness claims always state scope, mechanism, uncertainty, and source health.

Logical interfaces

CMP-11-IF-01 / Command / Issue Lease

Command `Issue Lease` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-11-IF-02 / Command / Renew Lease

Command `Renew Lease` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-11-IF-03 / Command / Record Logical Time

Command `Record Logical Time` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-11-IF-04 / Command / Declare Time Source Degraded

Command `Declare Time Source Degraded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-11-IF-05 / Query / Get Current Time Context

Query `Get Current Time Context` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-11-IF-06 / Query / Compare Causality

Query `Compare Causality` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-11-IF-07 / Query / Get Freshness

Query `Get Freshness` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-11-IF-08 / Event / Lease Issued

Event `Lease Issued` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-11-IF-09 / Event / Lease Expired

Event `Lease Expired` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-11-IF-10 / Event / Time Source Degraded

Event `Time Source Degraded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Healthy > Degraded > Unsynchronized > Recovering > Healthy. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: clock regression, split-brain lease, expired authority use, causality ambiguity, time-source compromise. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-11-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-11-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-11-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-11-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-11-TST-005 - Critical invariant. Ordering and freshness claims always state scope, mechanism, uncertainty, and source health. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-11-TST-006 - Failure containment. Inject clock regression, split-brain lease, expired authority use, causality ambiguity, time-source compromise. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-11-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-11-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-11-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-11-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.